GIH THE SWEDISH SCHOOL OF SPORT AND HEALTH SCIENCES

# An Exploration of Kinematic Data Collection in Biomechanics: from Muybridge to Invasive to Markerless.

## Where is AI taking us now?





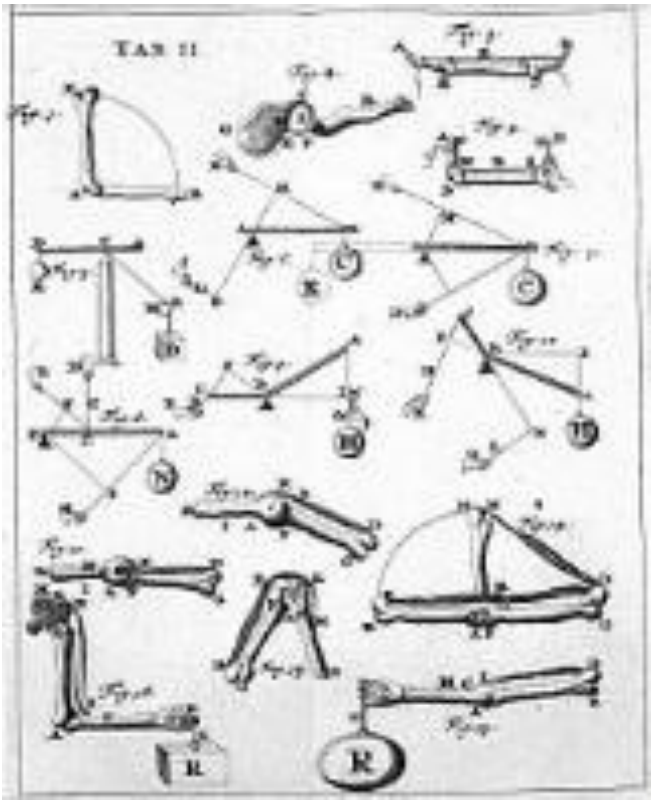
Toni Arndt

*toni.arndt@gih.se*

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# Giovanni Borelli (1608 – 1679)

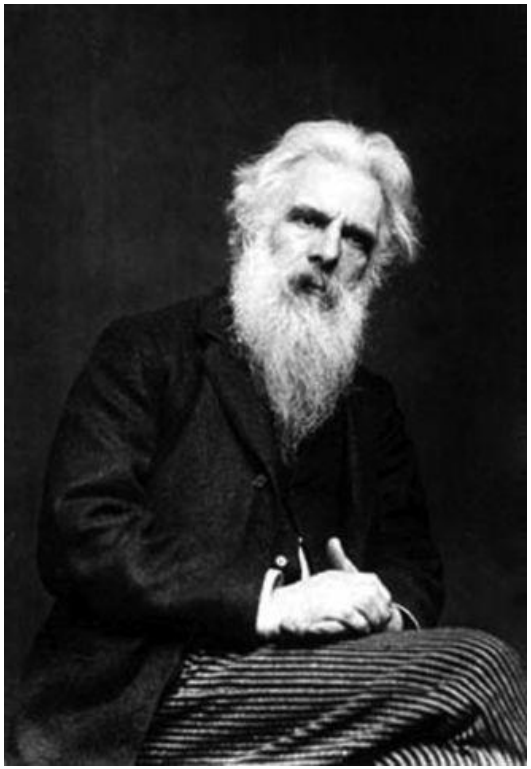




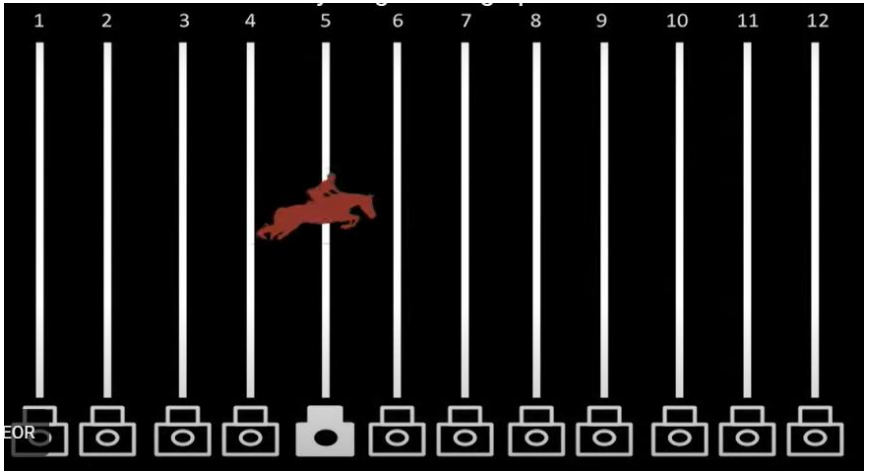

Giovanni Alfonso Borelli.
*De Motu Animalium, 1680, 1681*

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# Eadweard Muybridge (1830 – 1904)





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# Eadward Muybridge, June 19, 1878. Palo Alto



Eadweard Muybridge.
*Animal Locomotion 1887*

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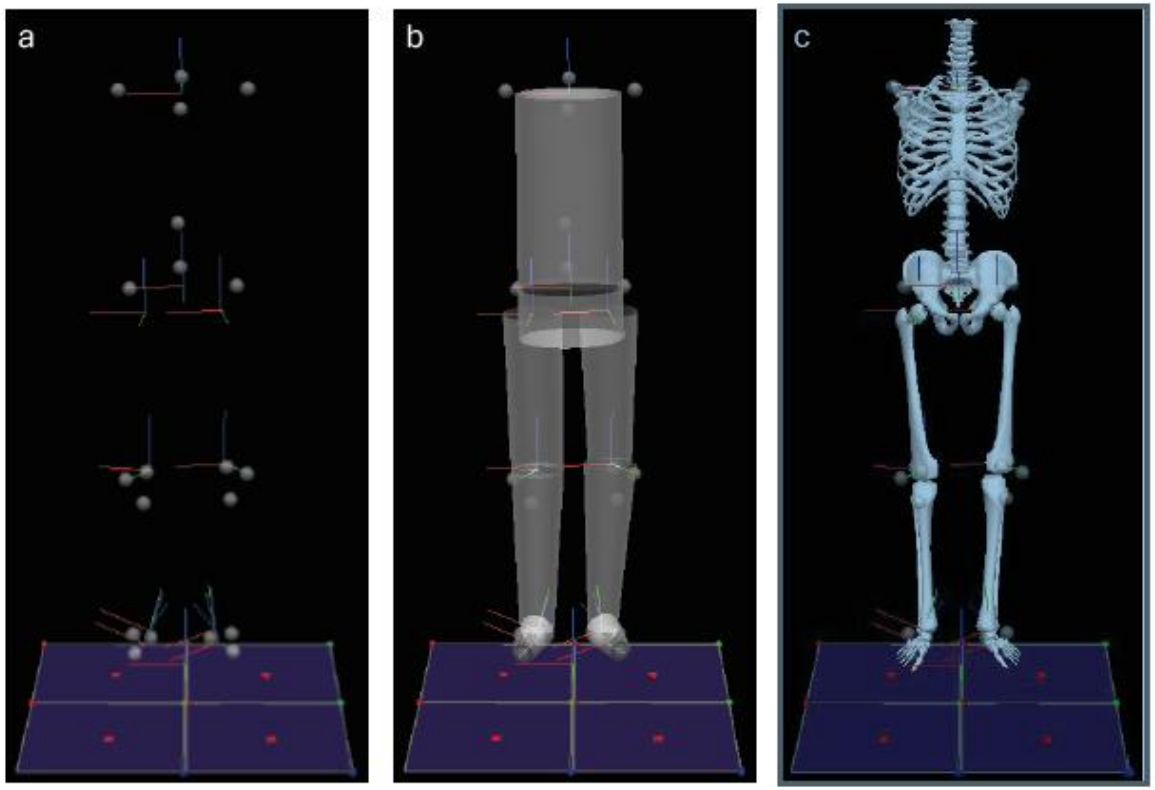
# Eadweard Muybridge



Eadweard Muybridge.
*Animal Locomotion 1887*

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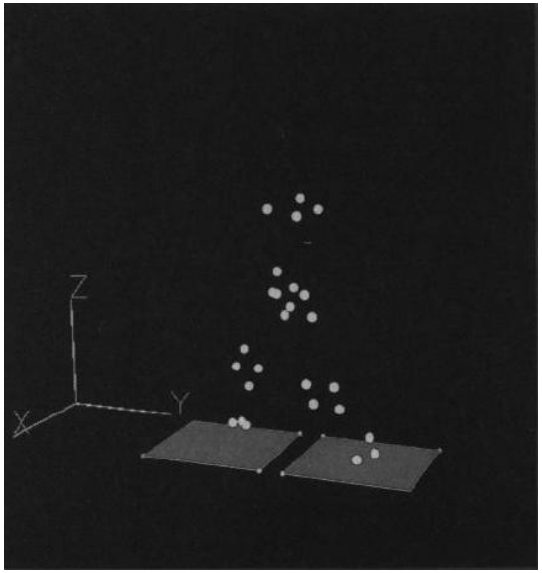
# Traditional marker-based motion capture





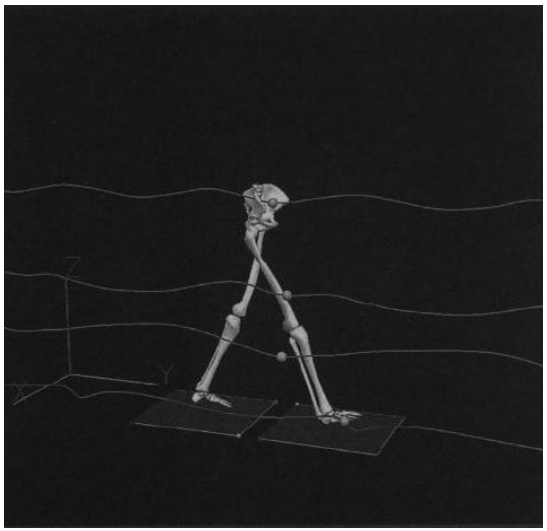

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# Traditional marker-based motion capture




+

- Highly precise 3D location of marker position
- High speed dynamic
- Non-invasive
- Well established / ”simple” use



-

- Time required for marker application
- Skin movement artefact

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# The foot

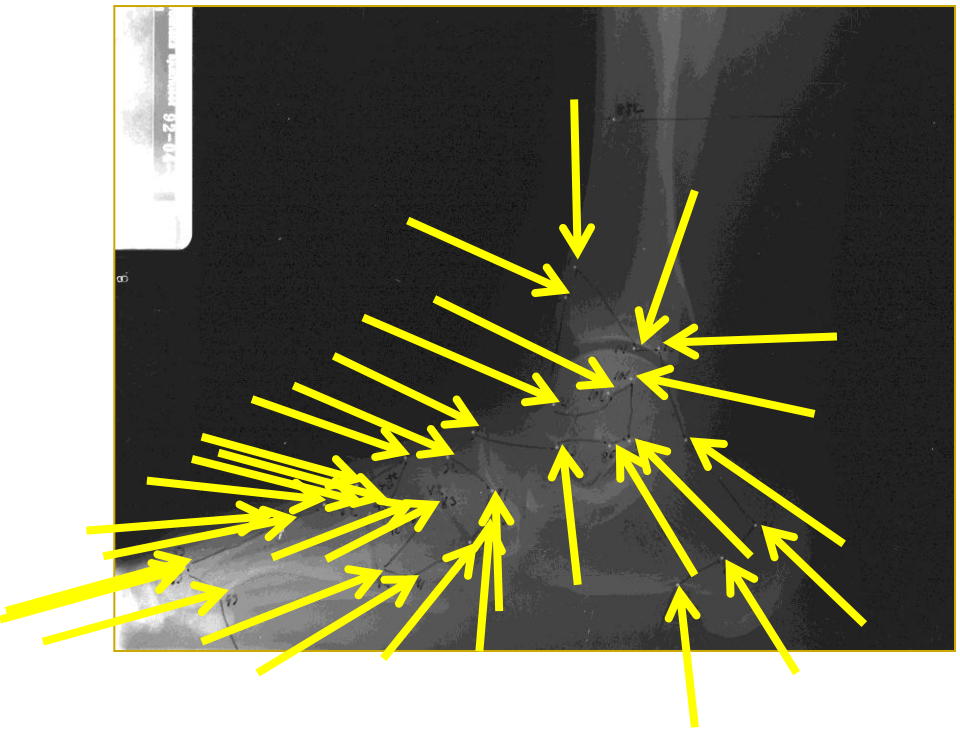
- Many small bones
- Few have bony landmarks on the skin surface
- Some do not even approach the skin surface (talus)
- Skin movement over many different bone segments
- Irregular geometry, joint surfaces, movements
- High accelerations

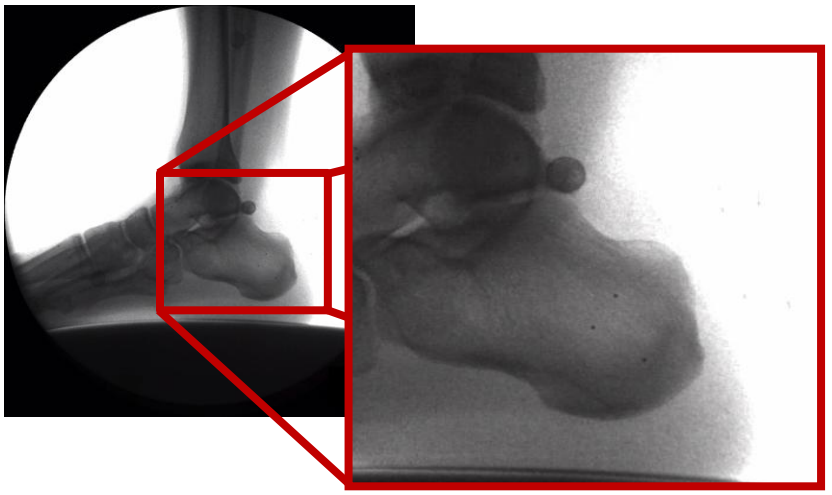




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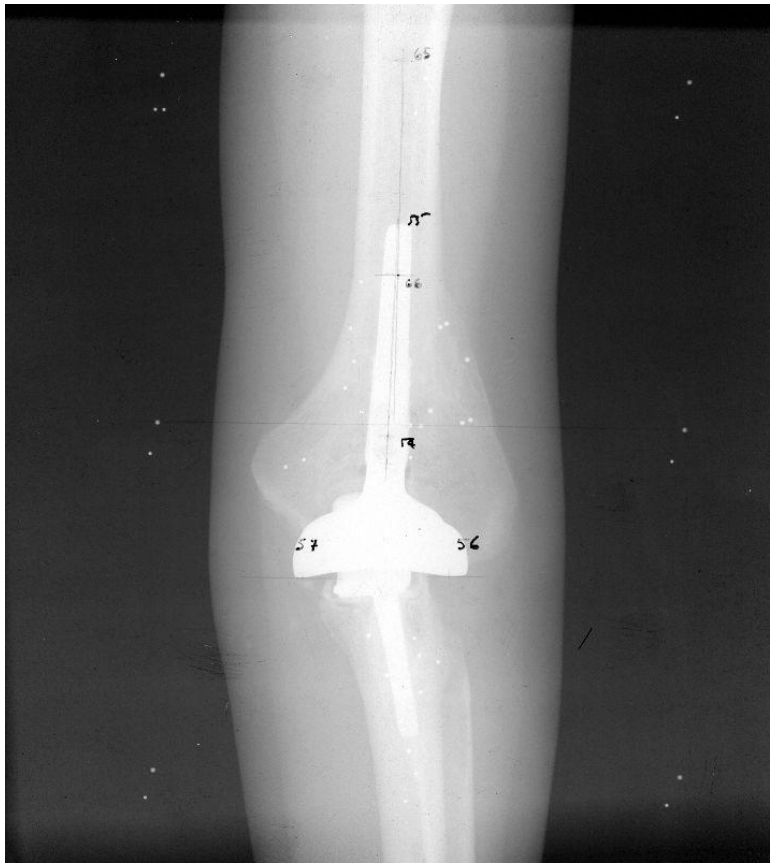
# Radiostereometric Analysis (RSA)

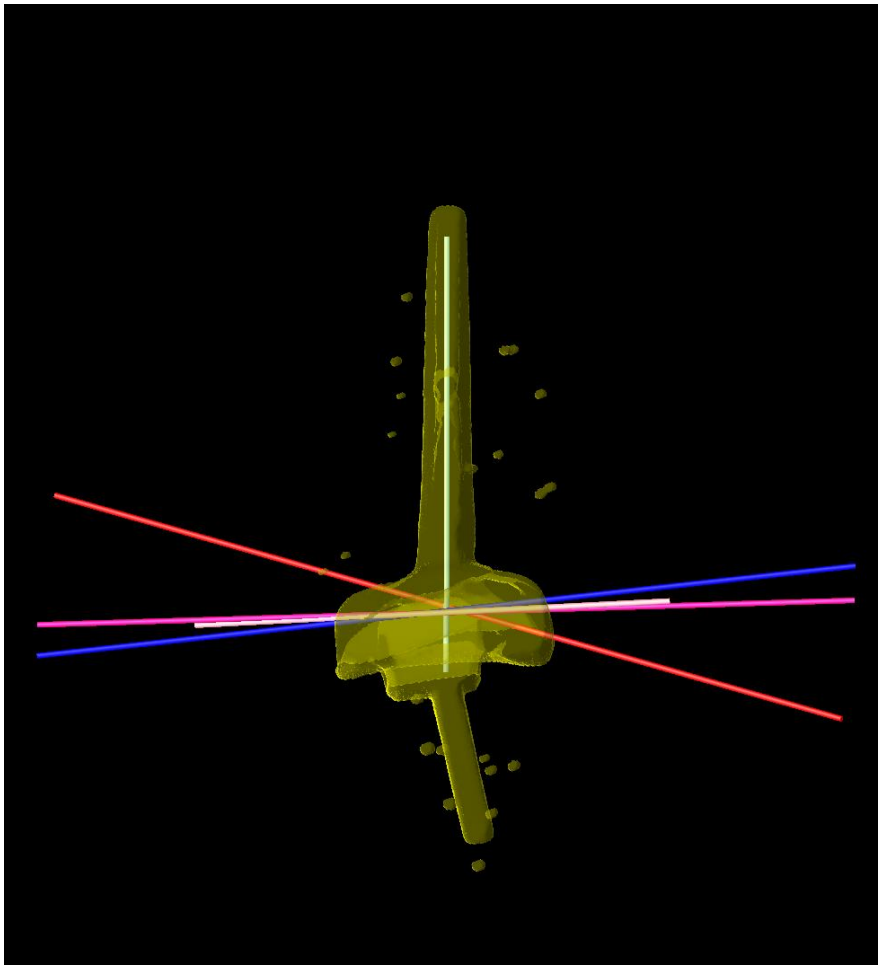




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# Radiostereometric Analysis (RSA)





Ericson *et al*. 2003. J Bone Joint Surg (Br)
Ericson *et al*. 2010. J Biomech

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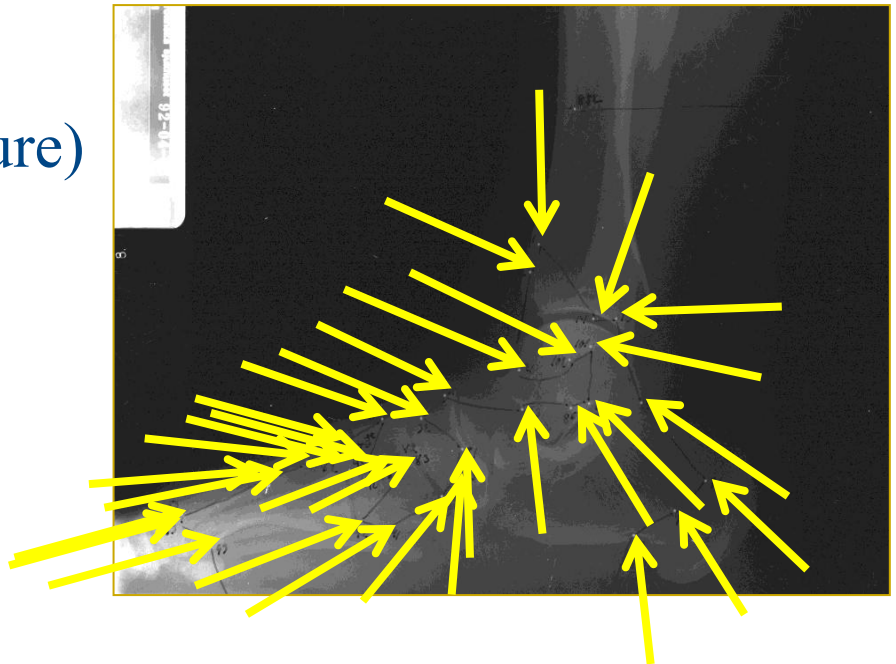
# Radiostereometric Analysis (RSA)

+

- Highly precise 3D location of marker position
- Direct marker location in bone segments
- Application to internal prosthesis motion
- Long-term follow-up

-

- Expensive (operational procedure)
- Invasive (short-term pain)
- Quasi-static
- Small n

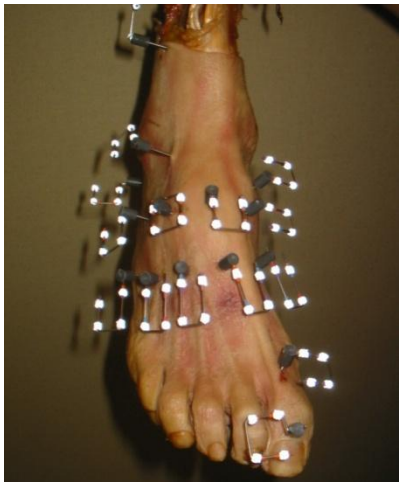


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# Cadaveric robot
















Nester *et al*. 2007. J Biomech

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# Intracortical pins

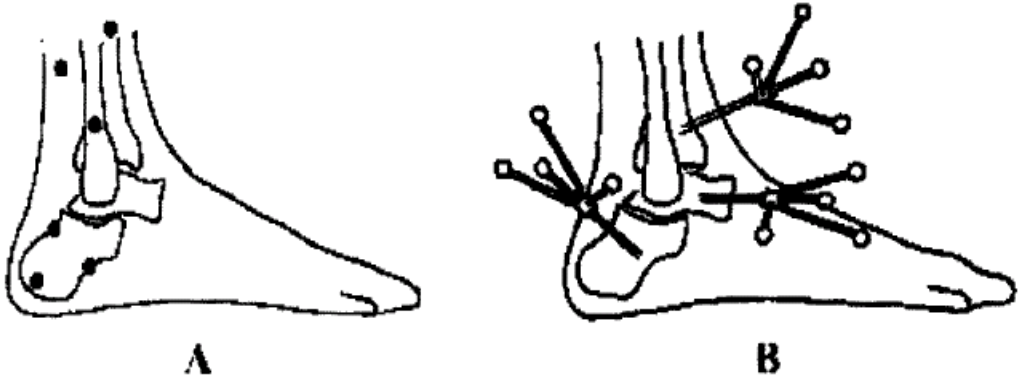


Biomechanics Laboratory Huddinge, Sweden 1994

Reinschmidt *et al*. (1997)

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# Intracortical pins






Westblad *et al*., 2002. Foot Ankle Int.

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# Intracortical pins

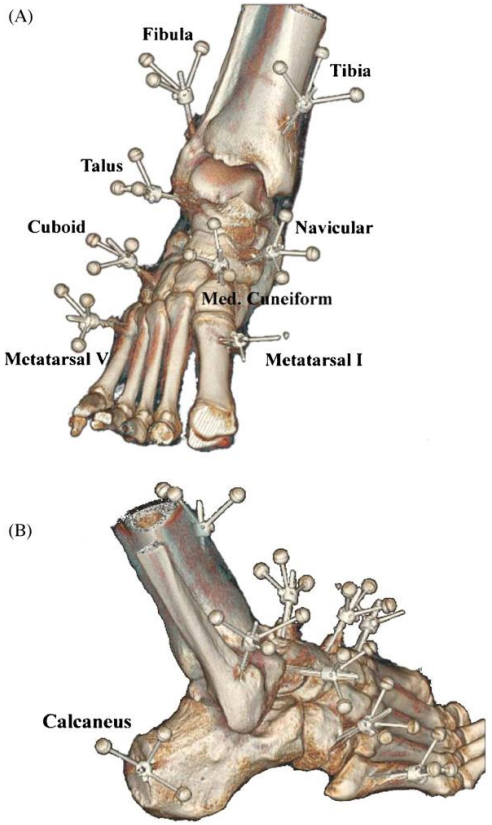


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# Intracortical pins







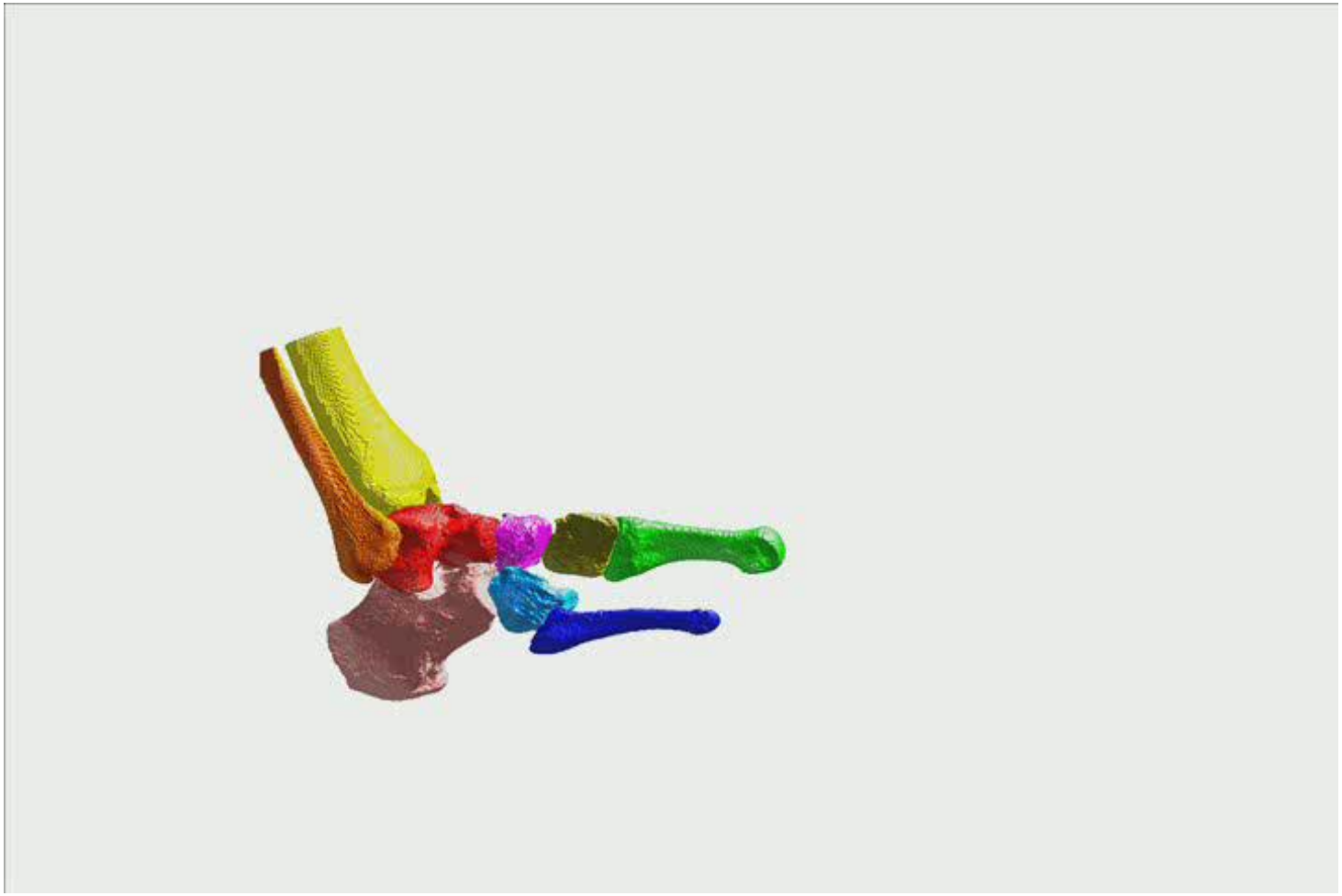

tibia

talus
calcaneus
navicular
med. cuneiform
metatarsal I
phalanx I / fibula

cuboid
metatarsal V

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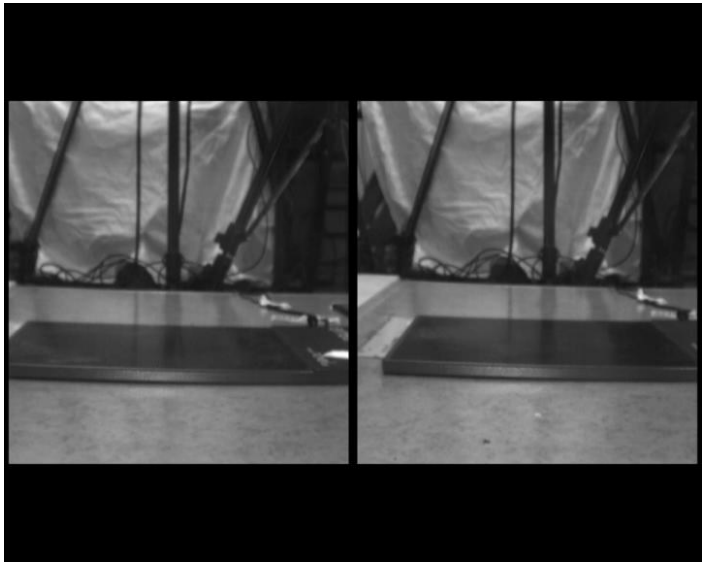
# Intracortical pins

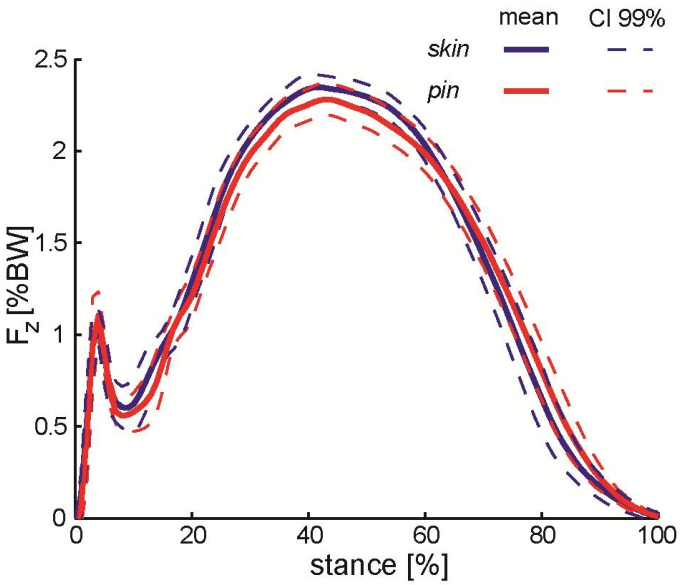


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# Intracortical pins

- Segment coordinate systems from standing trial, aligned with global coordinate system
- Coefficients of multiple correlation (CMC) GRF and tibia kinematics compared with and without bone pins (running)





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[redacted]

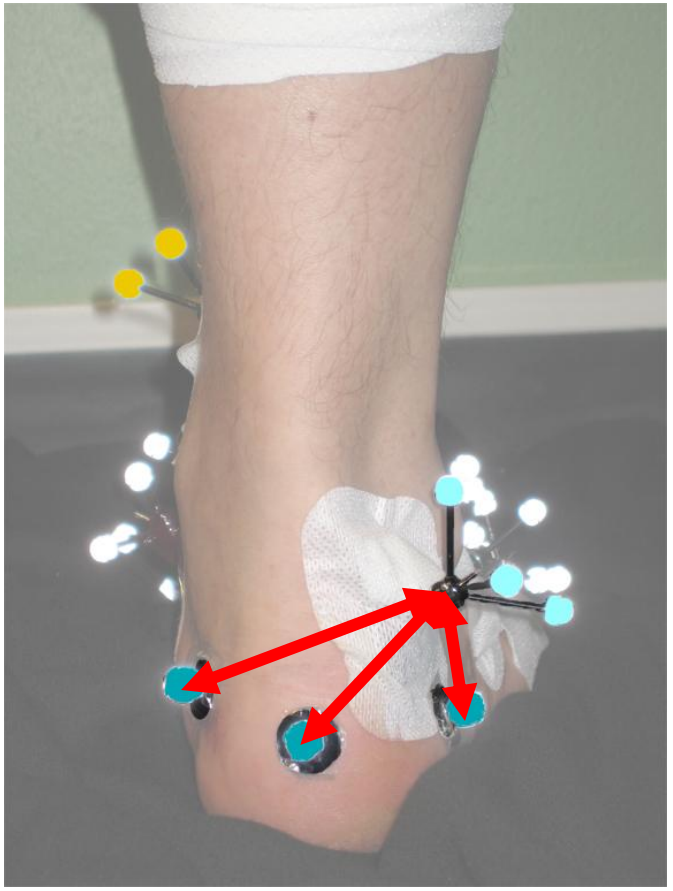
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[redacted]

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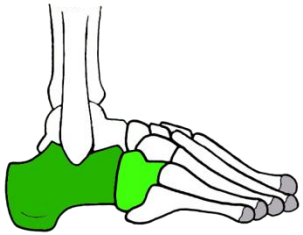
# Intracortical pins

| | phase | marker | $(\text{distance})_{max}$ |
|---|---|---|---|
| sagittal | 0…50% | lateral | 3 mm |
| | 51…100% | lateral | 4 mm |
| frontal | 0…50% | lateral | 4 mm |
| | 51…100% | | < 1 mm |
| transverse | 0…50% | medial | 2 mm |
| | 51…100% | | < 1 mm |



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# Walking ROM



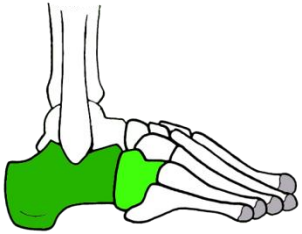




| | Mean (SD) | Mean (SD) | Mean (SD) | Mean (SD) | |
|---|---|---|---|---|---|
| sagittal | **9.7 (5.2)** | **7.2 (2.4)** | **11.5 (1.8)** | **5.3 (2.0)** | **13.3 (1.4)** |
| frontal | **11.3 (3.9)** | **8.8 (4.4)** | **10.4 (6.3)** | **5.4 (1.0)** | **10.4 (3.7)** |
| transversal | **8.1 (2.0)** | **8.9 (4.3)** | **6.2 (4.2)** | **6.1 (1.1)** | **9.8 (2.1)** |

Lundgren *et al*. 2008. Gait Posture

# Running ROM







| | Mean (SD) | Mean (SD) | Mean (SD) | Mean (SD) | |
|---|---|---|---|---|---|
| sagittal | **7.8 (1.2)** | **5.5 (2.0)** | **7.1 (4.0)** | **4.9 (2.5)** | **11.4 (1.6)** |
| frontal | **6.3 (1.3)** | **5.5 (2.0)** | **8.1 (2.6)** | **5.3 (1.9)** | **5.1 (0.6)** |
| transversal | **6.9 (3.3)** | **5.6 (1.8)** | **4.1 (1.1)** | **4.3 (1.4)** | **9.6 (2.4)** |

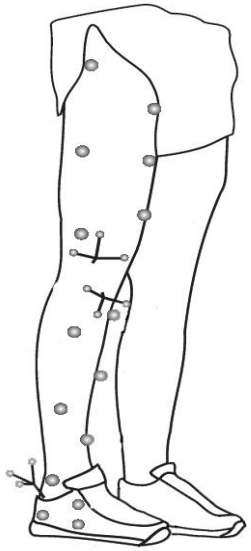
Arndt *et al*. 2007. J Biomech

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# Intacortical pins

+

- Highly precise 3D location of marker position
- Direct marker location in bone segments
- Dynamic motion





-

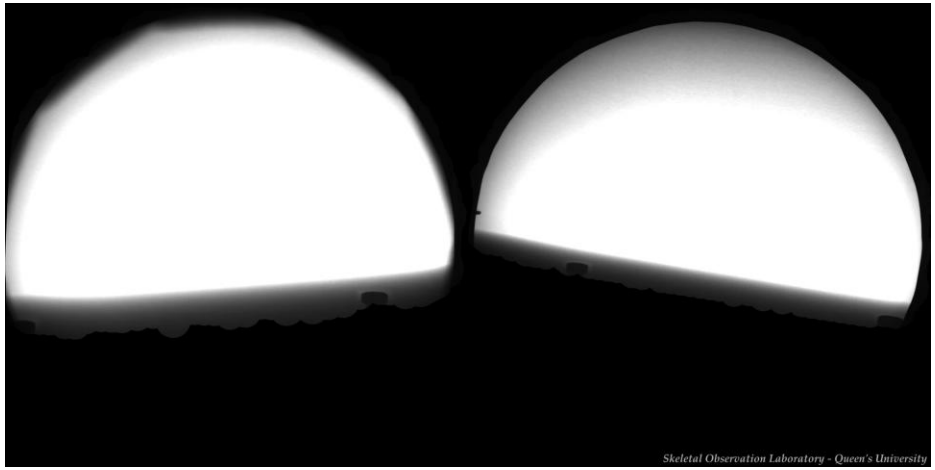
- Expensive (operational procedure)
- Invasive
- Small n
- Limited applied applications



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# Biplanar videoradiography

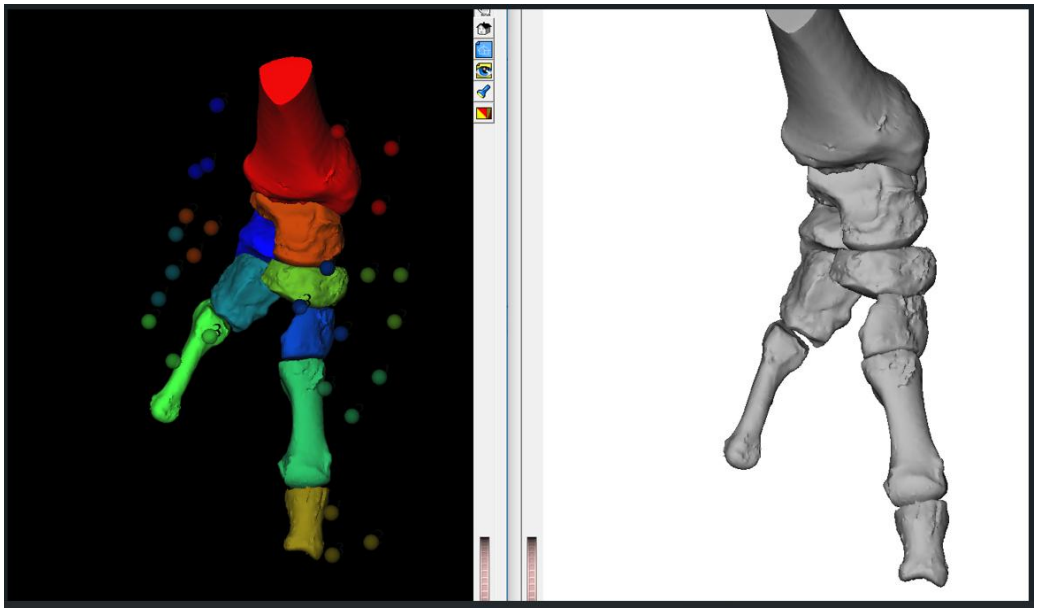


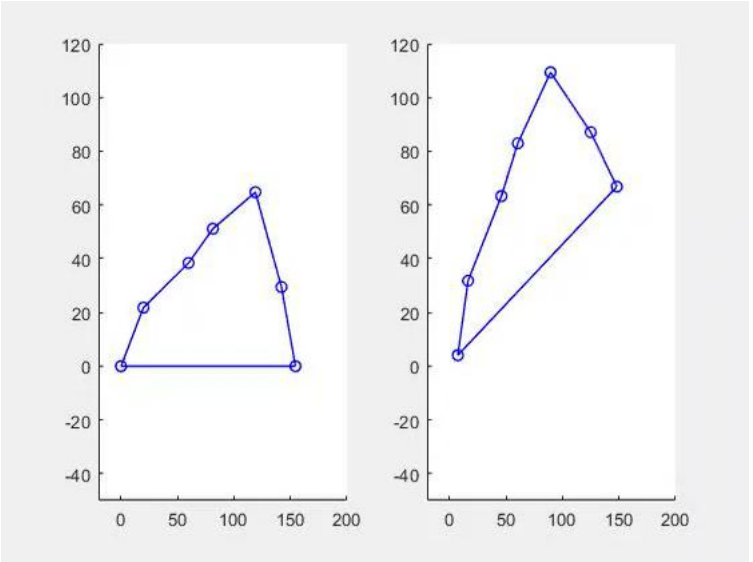


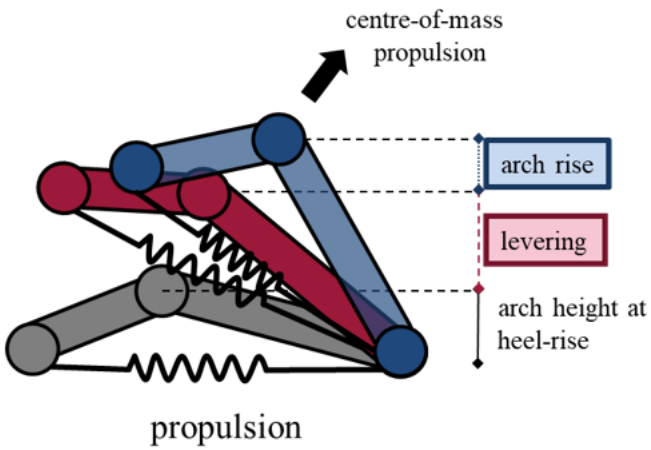


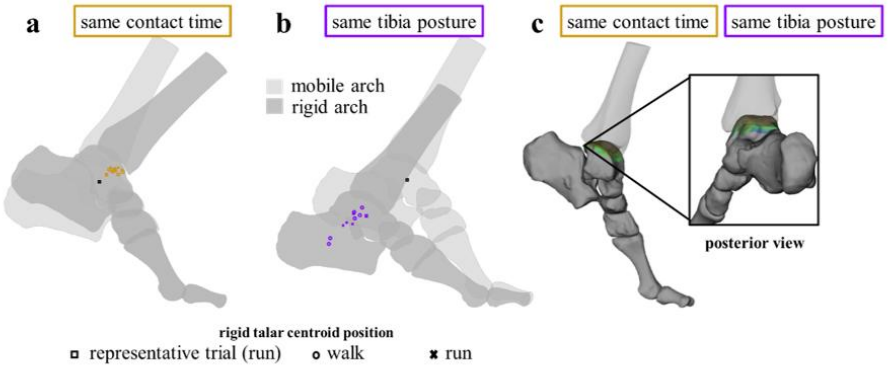

Michael Rainbow Laboratory, Queens University, Canada

# Biplanar videoradiography





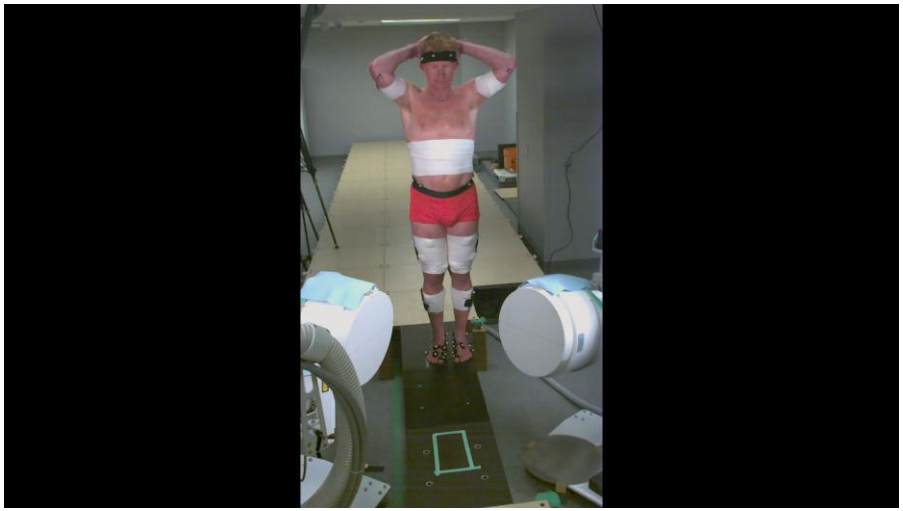



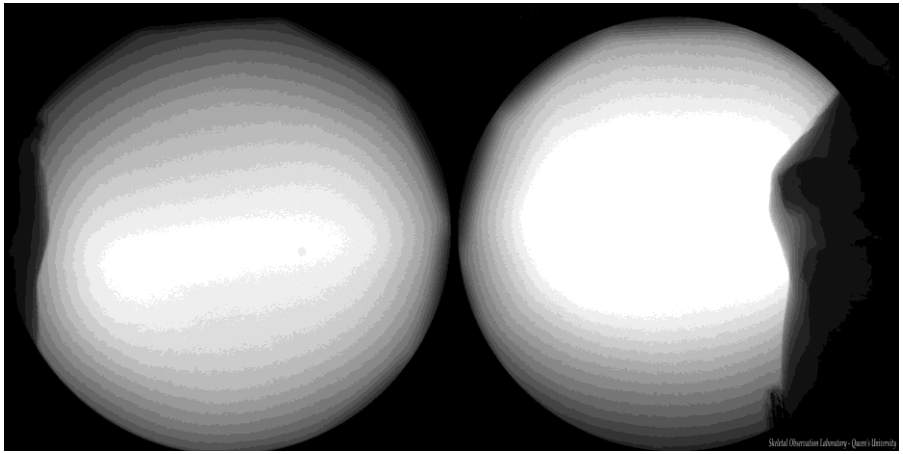




Welte *et al*. 2023. Front Bioeng Biotech

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# Biplanar videoradiography

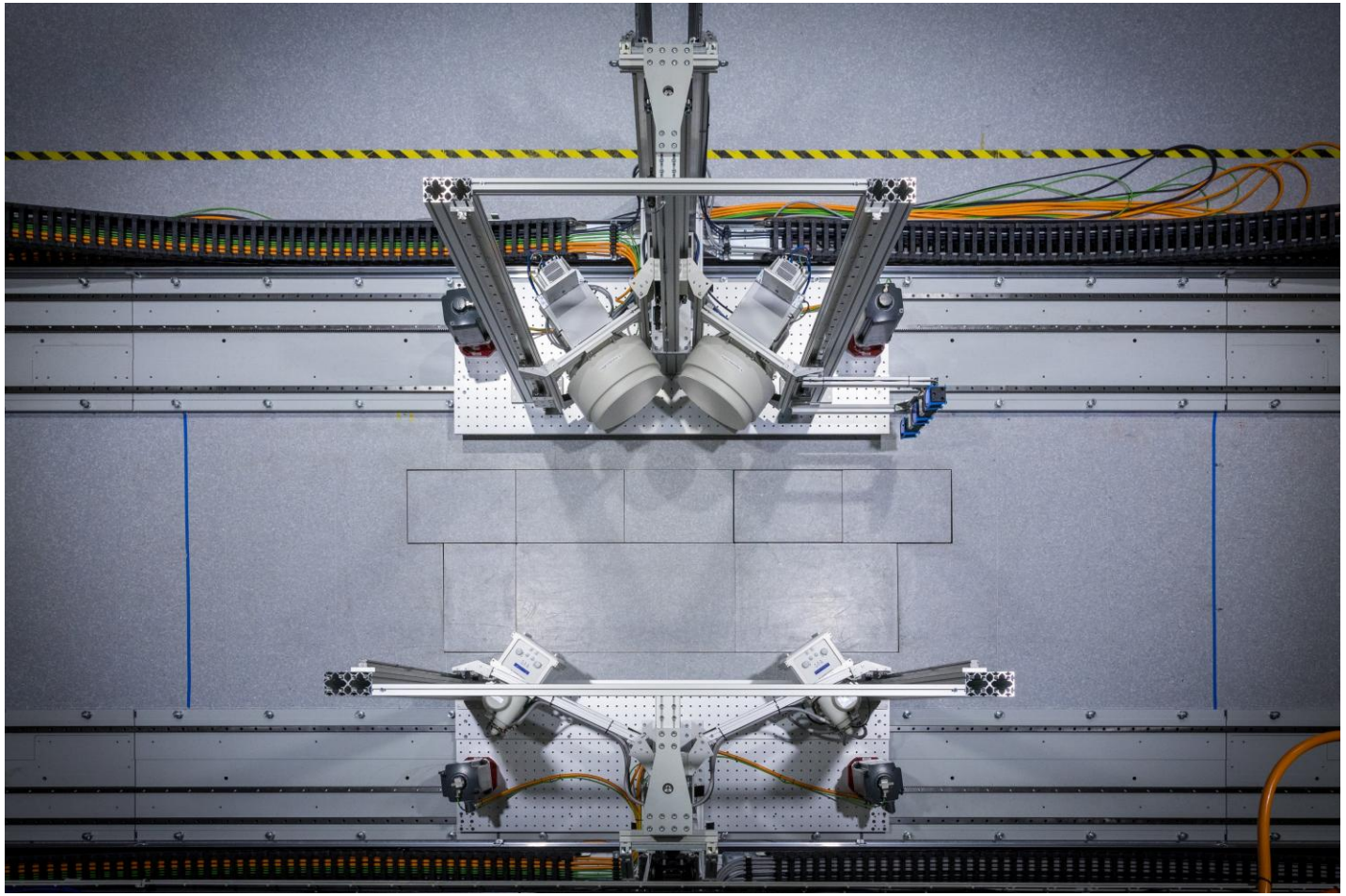




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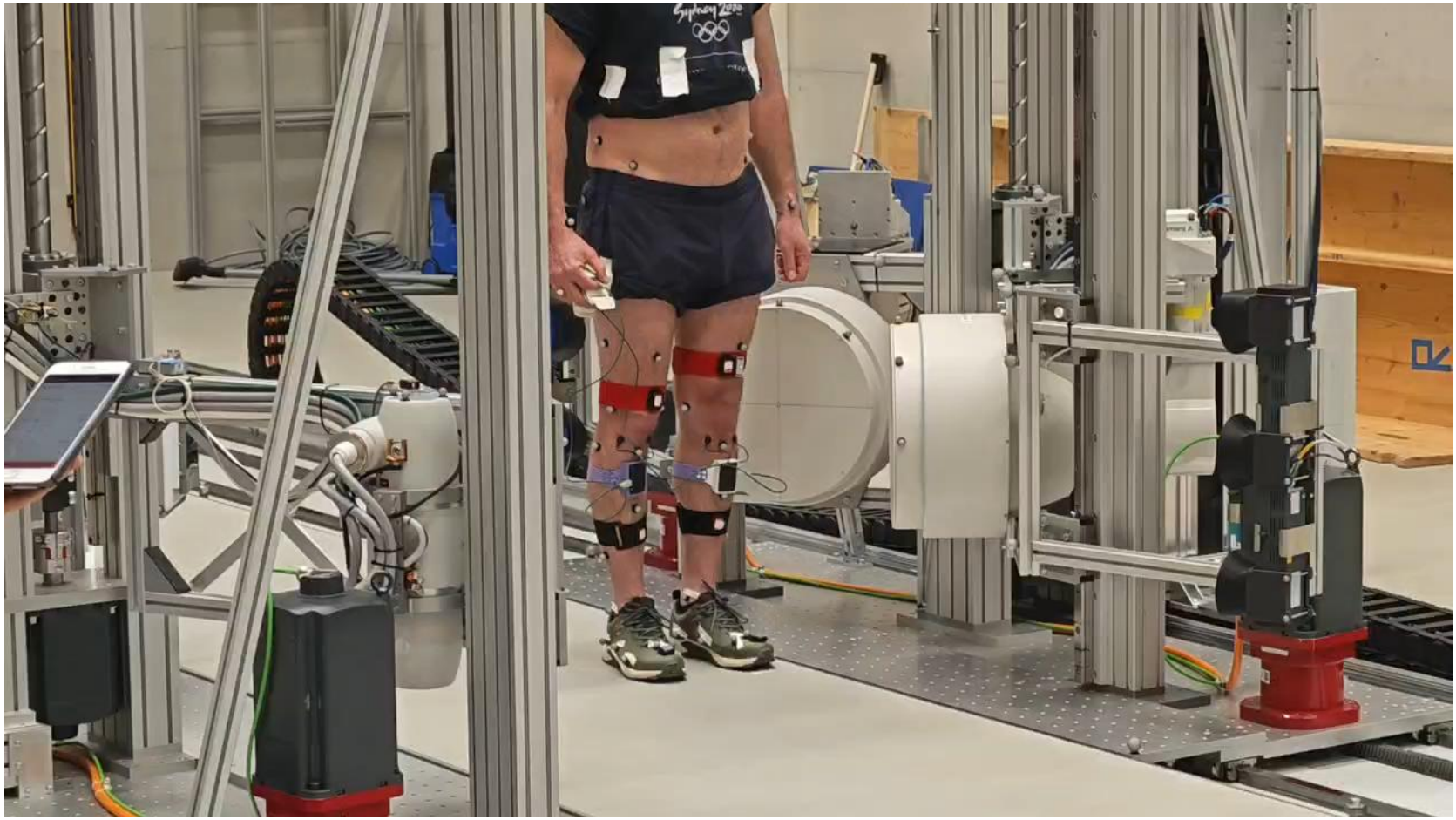
# Biplanar videoradiography



Bill Taylor Laboratory, ETH Zürich, Switzerland

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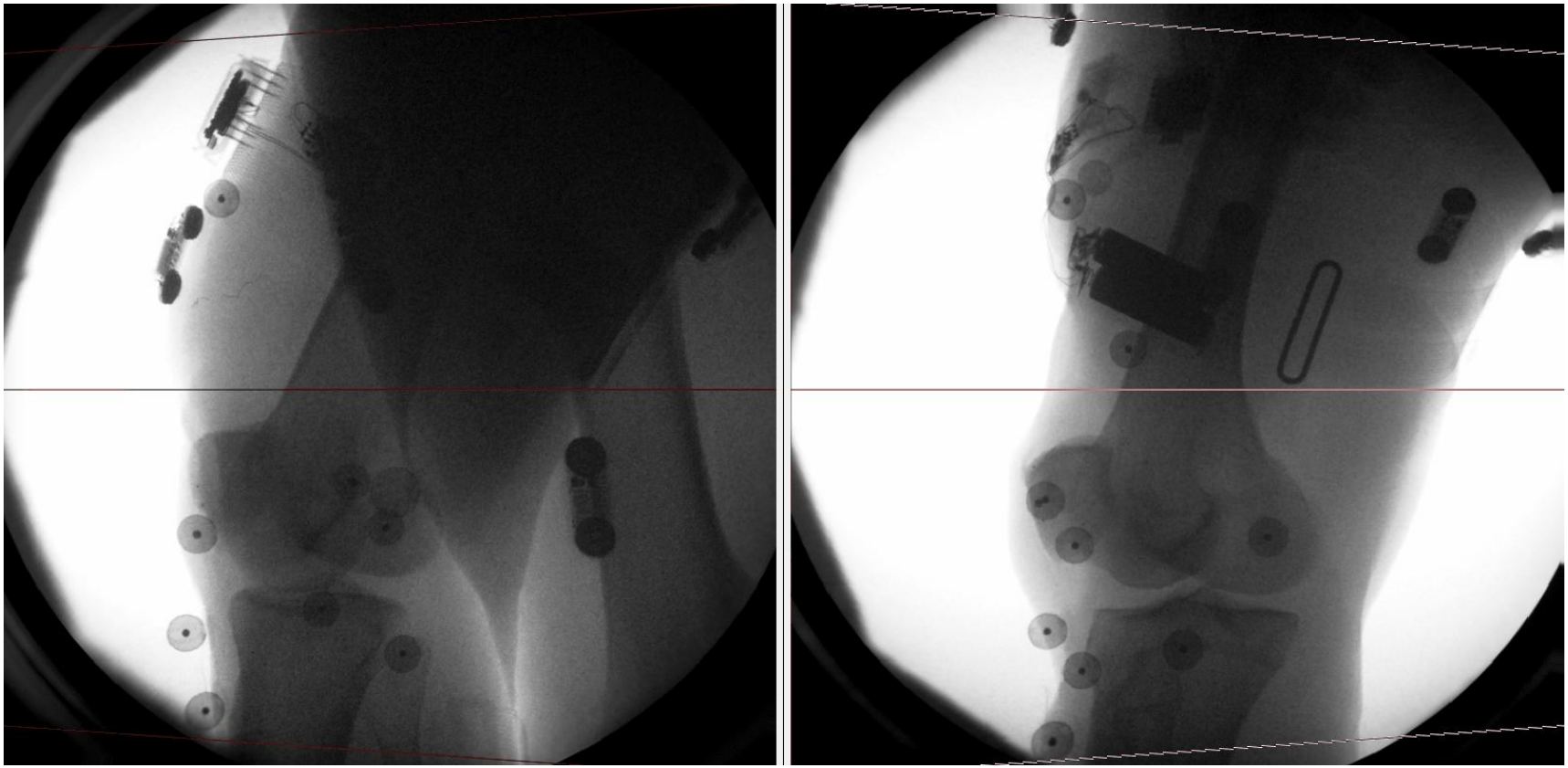
# Biplanar videoradiography



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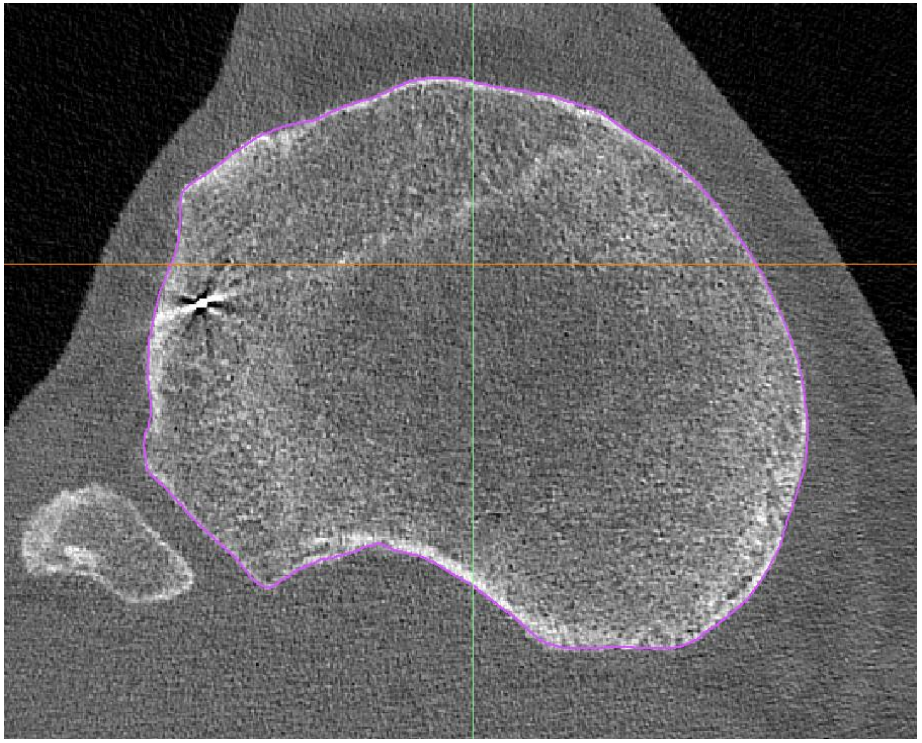
# Biplanar videoradiography



Bill Taylor Laboratory, ETH Zürich, Switzerland

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# Biplanar videoradiography



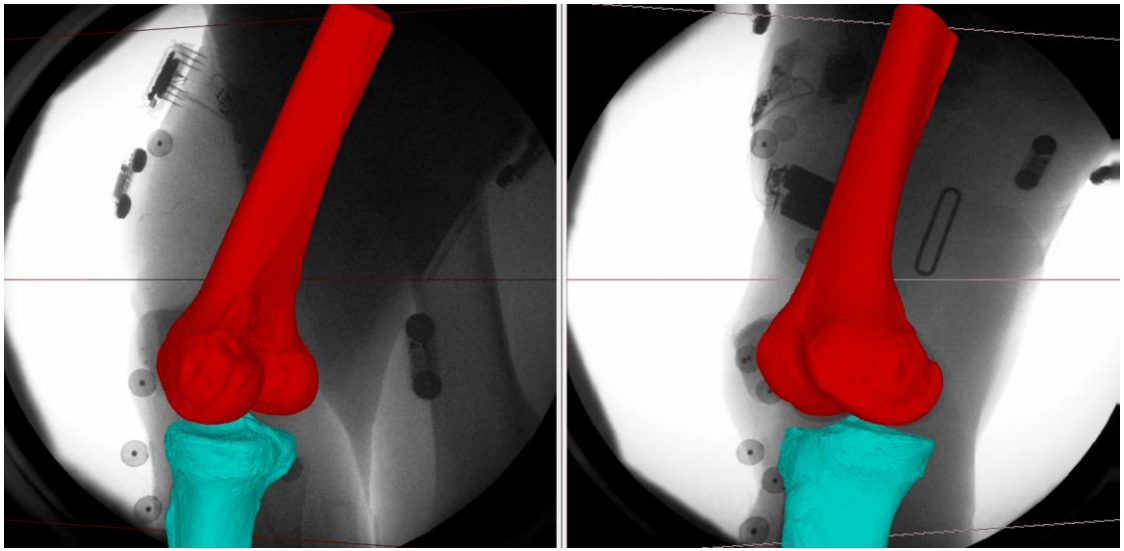


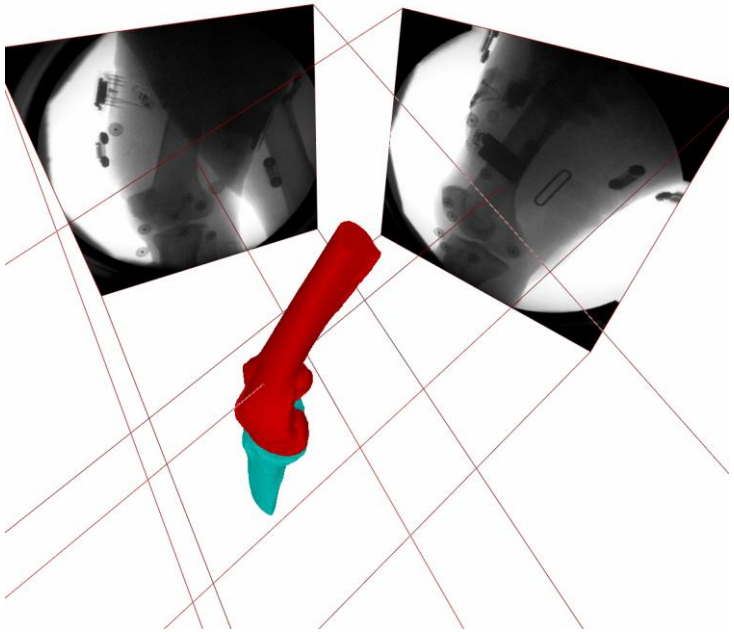
Bill Taylor Laboratory, ETH Zürich, Switzerland

ETH zürich

LABORATORY FOR MOVEMENT BIOMECHANICS

# Biplanar videoradiography





Bill Taylor Laboratory, ETH Zürich, Switzerland

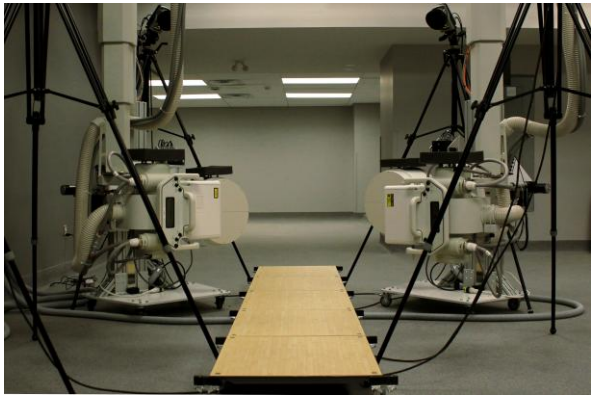
# Biplanar videoradiography

+

- Direct identification of bone segments
- Dynamic
- Non-invasive

-

- Very expensive and complex
- Dependent upon complex shape determination techniques
- Radiation exposure

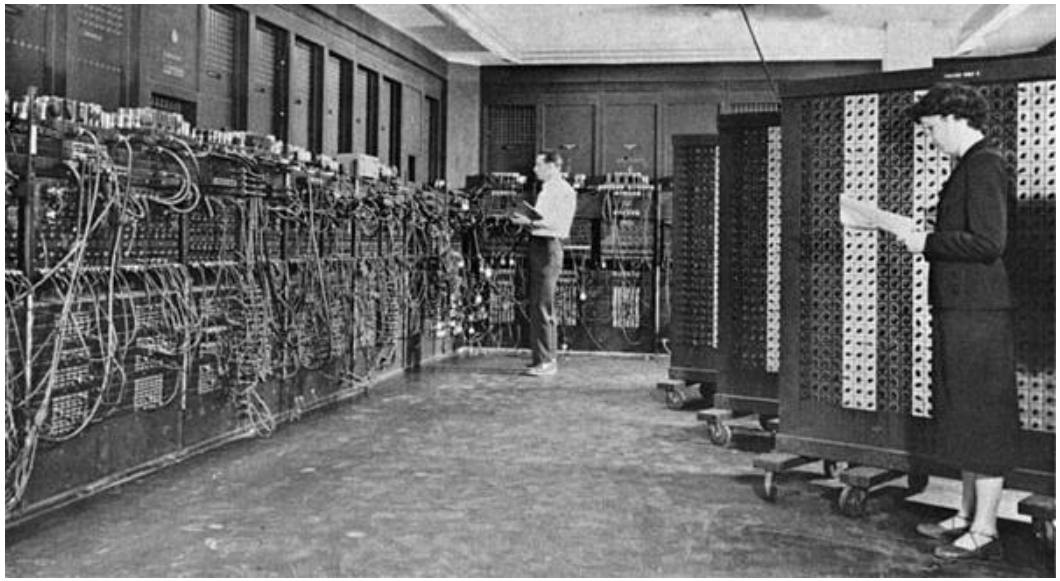


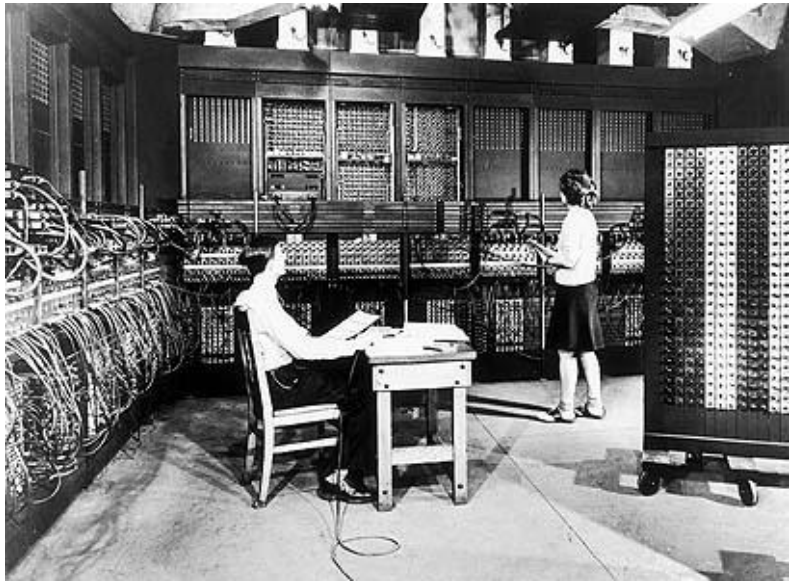


Michael Rainbow Laboratory,
Queens University, Canada

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# AI markerless motion analysis

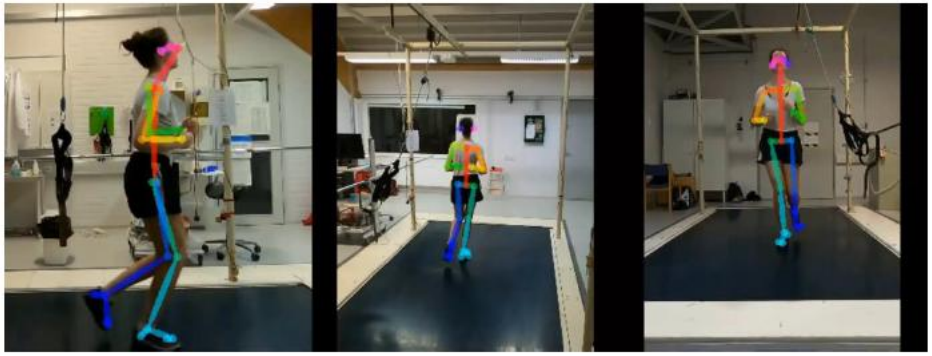




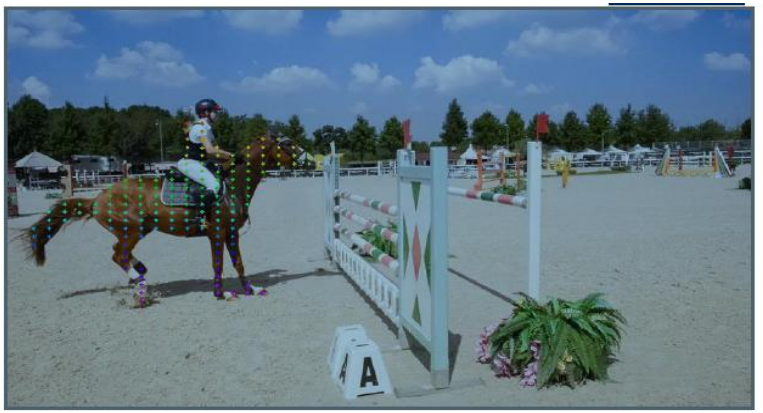


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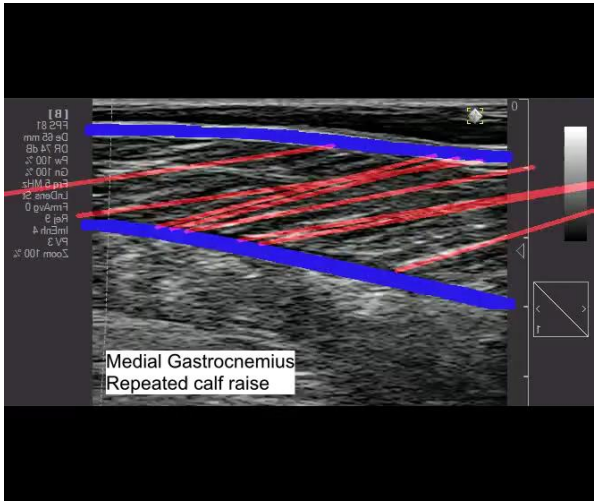
# AI markerless motion analysis



Cronin *et al*., 2020. J Biomech



Karaev *et al*., 2023







Neil Cronin, University of Jyväskylä, Finland

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# AI markerless motion analysis





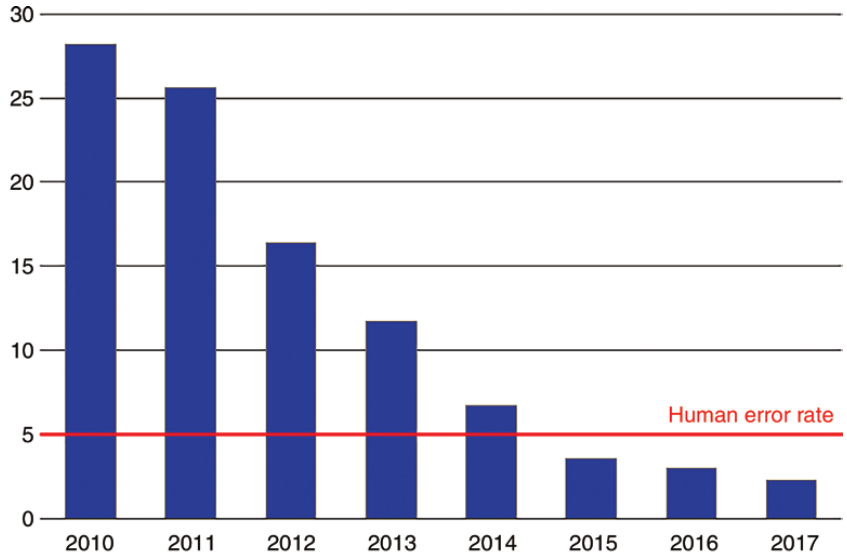


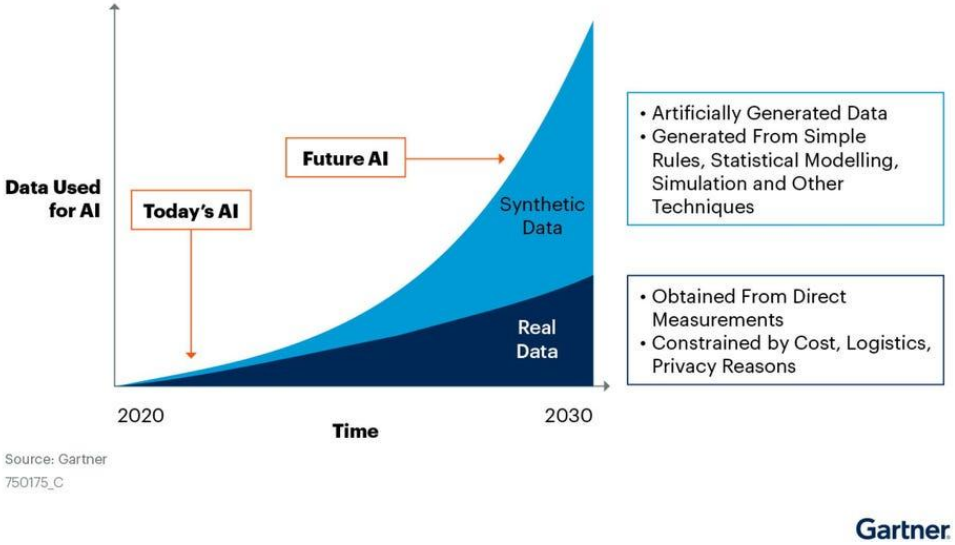
Patrick Almström, Qualisys, Sweden

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# AI markerless motion analysis

ImageNet Visual Recognition Error Rate (%)



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# AI markerless motion analysis

+

- Fast data collection
- Dynamic
- Non-invasive
- Large n and multiple people/objects



-

- Dependent upon big-data
- Who labels the data?
- Can it actually replace ”ground truth” skeletal data?









Laboratory for Movement Biomechanics






UNIVERSITY OF JYVÄSKYLÄ



















FACULTY OF KINESIOLOGY

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# Thank you